95%

1.) ⓐ Table 5E
$T = 328°F$ saturation

$u = u_f + x u_{fg}$

$500 = 298.19 + x(807.29)$ $x = 0.25$

$v = x v_{fg} + v_f = 0.01774 + 0.25(4.4327 - 0.1774)$

$= 1.12\ ft^3/lbm$

$h = h_f + x h_{fg} = 298.51 + 0.25(889)$

$= 521\ BTU/lbm$

ⓑ $P = 73.4\ psi$ saturation

$v = 0.032 + 0.5(3.9630 - 0.0032)$

$= 2.0\ ft^3/lbm$

$u = u_f + x u_{fg} = 327\ BTU/lbm$

ⓒ $v = 0.04034\ m^3/kg$

$T = 480°C$ $u = 3030\ kJ/kg$

$h = 3350\ kJ/kg$

superheated state

ⓓ $T = -20°C$ $h = 235.31\ kJ/kg$

$P = 133\ kPa$ $v = 0.147\ m^3/kg$

$u = 214\ kJ/kg$

saturated vapor state

v_g, V

V_1, V_2 2.) ⓐ saturated vapor $T_1 = 180°C$

$P_1 = 1002.8\ kPa$

ⓑ $v_1 = 0.19384\ m^3/kg$

$0.19384 = 0.001043 + x(1.6720 - 0.001043)$

$x = 0.11538$

3.) $m = 2\ kg$ $P_1 = 300\ kPa\ (3\ bar)$ $V_2 = 2.064\ m^3 = 72.889\ ft^3$

$v_{1g} = 1.5435\ ft^3/lbm \cdot 4.409\ lbm$ $2\ kg = 4.409\ lbm$

$V_1 = 6.805\ ft^3$ $T_1 = 417.35°F = 214.08°C$

$v_2 = \frac{V_2}{m} = \frac{72.889\ ft^3}{4.409\ lbm} = 16.53\ ft^3/lbm$

$v_2 > v_1$ water vapor is superheated

$\Rightarrow T_2 = 1283.65°F = 695.36°C$

$Work = P(V_2 - V_1)$

$= 300\ kPa = 43.51\ lbf/in^2\ (72.889 - 6.805) \cdot 1728\ in^3$

$= 3532850.64\ lbf \cdot in\ /\ 9338.03 = 3783.3\ BTU$

$= \boxed{3991.6\ kJ}$

80%

4.) $P_1 = 10$ bar $T_1 = 60°C$ $P_2 = 2$ bar $T_2 = 30°C$ $m = 1$ kg $w = 20$ kJ

10 bar @ $T = 60°C$

a) $v_1 = 0.02860$ m³/kg

b) $u_1 = 267.35$ kJ/kg

$u_2 = 253.06$ kJ/kg

$Q = w + m(u_2 - u_1)$

$Q = 20\text{kJ} (1\text{kg}) (253.06 - 208.35)$

$\boxed{Q = -305.8 \text{ kJ}}$

80%

80%

5.) $v_2 = 0.89133$ m³/kg

$u_2 = 2528.9$ kJ/kg

$P_1 = 47.416$ kPa

$v_{f1} = 0.001029$ m³/kg

$v_{g1} = 3.4053$ m³/kg

$0.89133 = 0.001029 + x(3.4053 - 0.001029)$

$x_1 = 0.261$

$u_1 = 334.97 + 0.261(2146.4) = 896.35$ kJ/kg

$Q = 2528.9 - 896.35 = 1632.55$ kJ/kg $\cdot 1.5$ kg

$\boxed{Q = 2448.8 \text{ kJ}}$

6.) $v_f = 0.8352\times10^{-3}$ m³/kg $v_g = 0.0236$ m³/kg

$v = 0.8352\times10^{-3} + 0.6 (0.0236 - 0.8352\times10^{-3})$

$v = 0.01449$ m³/kg

$x = m_g / (m_g + m_f)$ $0.6 = \frac{m_g}{m_g + 25}$ $m_g = 37.5$ kg

$V_g = 0.0236 \times 37.5 = 0.885$ m³

$V = m \cdot v = (25 + 37.5) \cdot 0.01449 = \underline{0.9056 \text{ m}^3}$

$x_{vapor} = \frac{0.885}{0.9056} \cdot 100 = \underline{97.725}$

7.) $R_{He} = \frac{\bar{R}}{M} = \frac{8.314}{4} = 2.0785 \frac{\text{kJ}}{\text{kg K}}$

$V_1 = \frac{0.2 \cdot 10^{-3} \cdot 2.0785 \cdot 300}{1.3\times10^2}$

$V_1 = 0.959\times10^{-3}$ m³ $= \underline{959 \text{ cm}^3}$

$V_2 = \frac{0.2\times10^{-3} \cdot 2.0785 \cdot 290}{0.8\times10^2}$

$V_2 = 1.506\times10^{-3}$ m³

$= \underline{1506 \text{ cm}^3}$

8.) 25°C @ 2.75 bar c 0.05 m^3 2 57°C 2.75 kJ

298.15 275000 Pa 530.15 °K

$R_{CO_2} = 0.189\ kJ/kg \cdot K$

$m = \frac{275000 \cdot 0.05}{0.189 \times 10^3 \cdot 298.15} = \boxed{0.244\ kg}$

$w = 2.75\ kJ - n(u_2 - u_1)$

$n = \frac{m}{mw} = \frac{0.244\ kg}{44.01} \cdot 1000 = 5.544\ mol$

298.15 K = $u_1 = 6.95$ kJ/mol

530.15 K $u_2 = 14.7$ kJ/mol

$w = 2.75\ kJ - 5.544(14.7 - 6.95)$

$\boxed{w = -40.216\ kJ}$